University Of Chenab, Gujrat

Course: Software Testing and Quality Assurance

(Spring 2026)

Assignment 01

Basis Path Testing and Cyclomatic Complexity Analysis

Java implementation, flow graph analysis, and test case design

Prepared in response to the assignment brief that required a set of test cases, basis path testing, and cyclomatic complexity for three problem scenarios.

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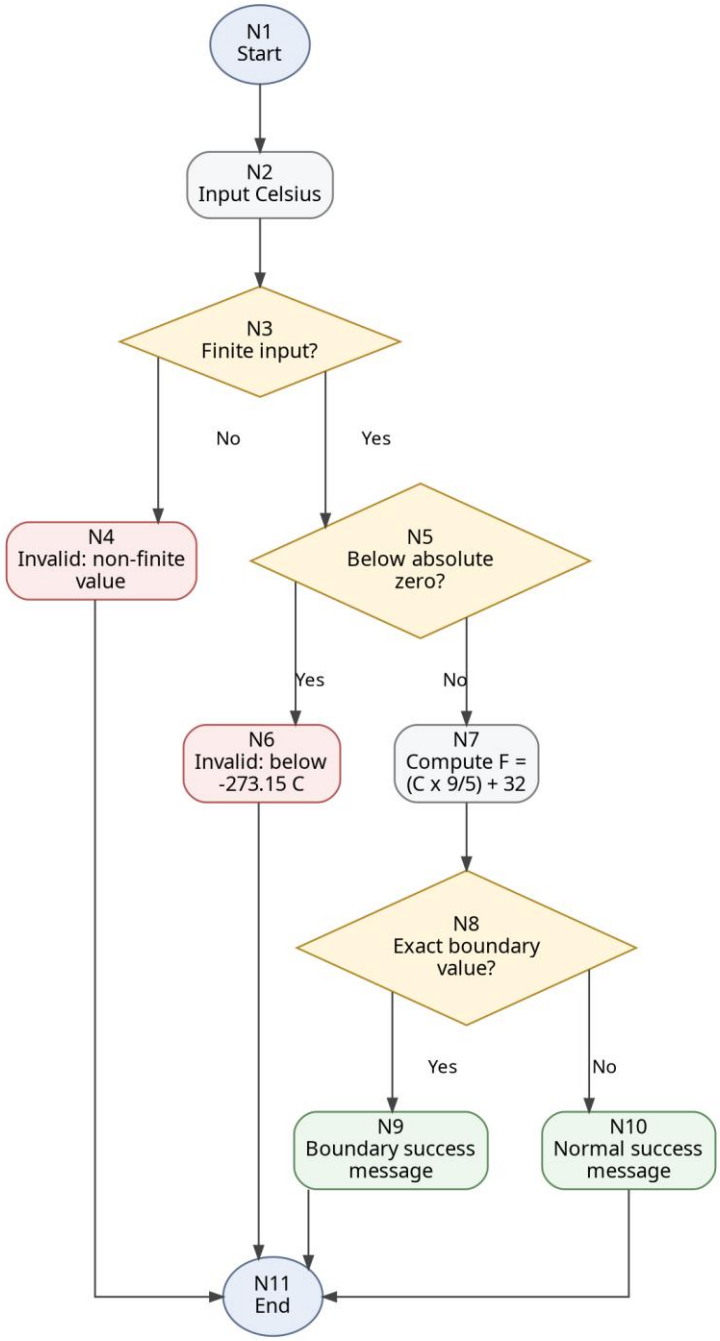
1. Celsius to Fahrenheit Converter

Flow Graph



Cyclomatic Complexity

Predicate nodes = 3

$V(G) = D + 1 = 3 + 1 = 4$

$V(G) = E - N + 2P = 13 - 11 + 2(1) = 4$

Basis Paths

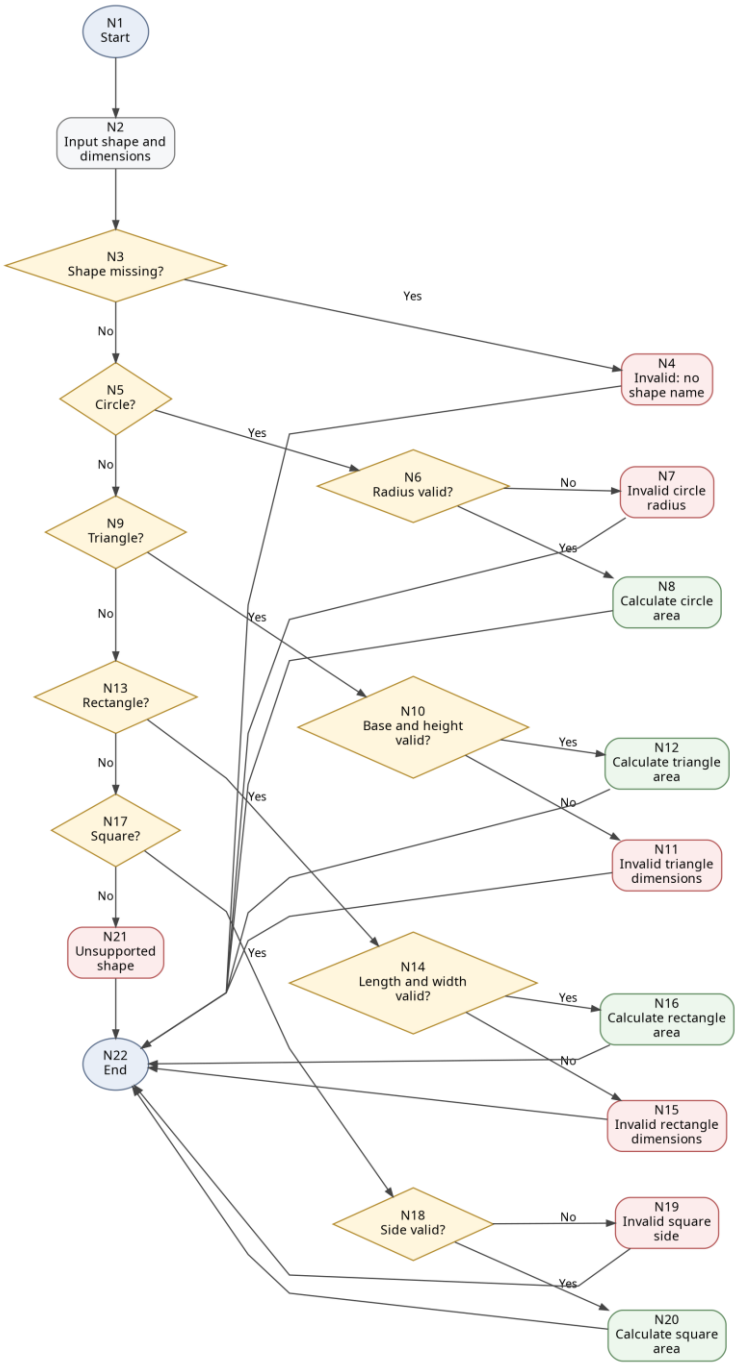
Path	Node Sequence
P1	N1 -> N2 -> N3(No) -> N4 -> N11
P2	N1 -> N2 -> N3(Yes) -> N5(Yes) -> N6 -> N11
P3	N1 -> N2 -> N3(Yes) -> N5(No) -> N7 -> N8(Yes) -> N9 -> N11
P4	N1 -> N2 -> N3(Yes) -> N5(No) -> N7 -> N8(No) -> N10 -> N11

Test Cases

Test Case	Input	Path	Expected Result	Status
TC-CF-01	C = NaN	P1	Validation message for non-finite input	Pass
TC-CF-02	C = -300.0	P2	Validation message for value below absolute zero	Pass
TC-CF-03	C = -273.15	P3	Fahrenheit = -459.67	Pass
TC-CF-04	C = 25.0	P4	Fahrenheit = 77.00	Pass

2. Area of Circle, Triangle, Rectangle, and Square Finder

Flow Graph



Cyclomatic Complexity

Predicate nodes = 9

V(G) = D + 1 = 9 + 1 = 10

V(G) = E - N + 2P = 30 - 22 + 2(1) = 10

Basis Paths

Path	Node Sequence
P1	N1 -> N2 -> N3(Yes) -> N4 -> N22
P2	N1 -> N2 -> N3(No) -> N5(Yes) -> N6(No) -> N7 -> N22
P3	N1 -> N2 -> N3(No) -> N5(Yes) -> N6(Yes) -> N8 -> N22
P4	N1 -> N2 -> N3(No) -> N5(No) -> N9(Yes) -> N10(No) -> N11 -> N22
P5	N1 -> N2 -> N3(No) -> N5(No) -> N9(Yes) -> N10(Yes) -> N12 -> N22
P6	N1 -> N2 -> N3(No) -> N5(No) -> N9(No) -> N13(Yes) -> N14(No) -> N15 -> N22
P7	N1 -> N2 -> N3(No) -> N5(No) -> N9(No) -> N13(Yes) -> N14(Yes) -> N16 -> N22
P8	N1 -> N2 -> N3(No) -> N5(No) -> N9(No) -> N13(No) -> N17(Yes) -> N18(No) -> N19 -> N22
P9	N1 -> N2 -> N3(No) -> N5(No) -> N9(No) -> N13(No) -> N17(Yes) -> N18(Yes) -> N20 -> N22
P10	N1 -> N2 -> N3(No) -> N5(No) -> N9(No) -> N13(No) -> N17(No) -> N21 -> N22

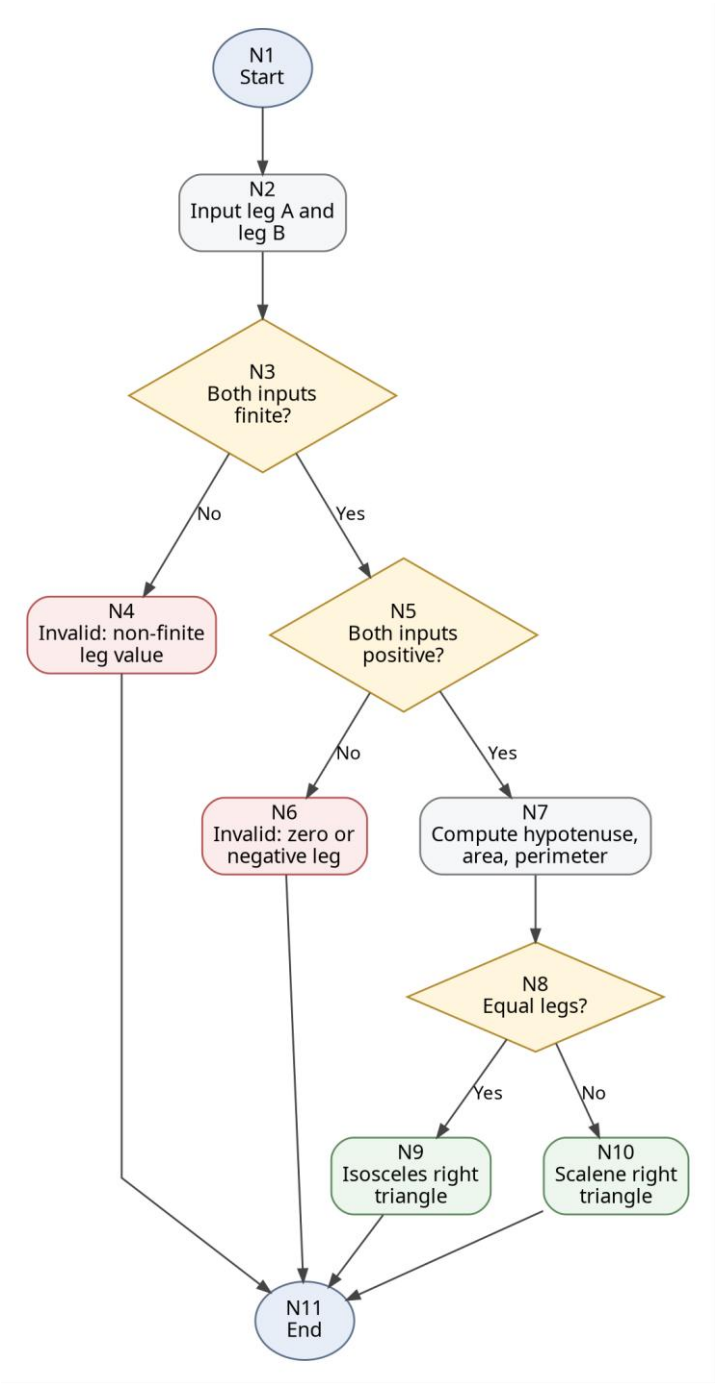
Test Cases

Test Case	Input	Path	Expected Result	Status
TC-AR-01	shape = ""	P1	Validation message for missing shape	Pass
TC-AR-02	circle, r = -2	P2	Invalid circle radius	Pass
TC-AR-03	circle, r = 7	P3	Area = 153.94 (approx.)	Pass
TC-AR-04	triangle, b = -3, h = 5	P4	Invalid triangle dimensions	Pass
TC-AR-05	triangle, b = 10, h = 4	P5	Area = 20.00	Pass

TC-AR-06	rectangle, l = 0, w = 8	P6	Invalid rectangle dimensions	Pass
TC-AR-07	rectangle, l = 6, w = 3	P7	Area = 18.00	Pass
TC-AR-08	square, s = -5	P8	Invalid square side	Pass
TC-AR-09	square, s = 9	P9	Area = 81.00	Pass
TC-AR-10	shape = pentagon	P10	Unsupported shape message	Pass

3. Right Angle Triangle Calculator

Flow Graph



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Basis Paths

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Test Cases

Test Case	Input	Path	Expected Result	Status
TC-RT-01	a = NaN, b = 4	P1	Validation message for non-finite input	Pass
TC-RT-02	a = 0, b = 4	P2	Validation message for non-positive leg	Pass
TC-RT-03	a = 5, b = 5	P3	Hypotenuse = 7.0711, area = 12.5, perimeter = 17.0711	Pass
TC-RT-04	a = 3, b = 4	P4	Hypotenuse = 5.0, area = 6.0, perimeter = 12.0	Pass